

Preface

This is the second course in a two-quarter long sequence covering functional analysis, Hilbert space theory, and partial differential equations. Most of the material is covered in Ward Cheney's *Analysis for Applied Mathematics*, Nicholas Young's *An Introduction to Hilbert Space*, and Lawrence C. Evans's *Partial Differential Equations*.

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Chapter 1

Preliminaries

For the remainder of this chapter: let $U \subset \mathbf{R}^n$ be an open set. An element $x \in \mathbf{R}^n$ will always be of the form $(x_1, \dots, x_n)$. The function $u : U \rightarrow \mathbf{R}$ will always denote the unknown function being solved for in a differential equation. We denote $u := u(x) = u(x_1, \dots, x_n)$.

§ 1.1. Basic Definitions

Definition 1.1.1. The unit vector in $\mathbf{R}^n$ is denoted by e^i , where it is 1 in the i^{th} slot and 0 otherwise.

Definition 1.1.2. The partial derivative of u with respect to x_i is:

$$u_{x_i} := \frac{\partial u}{\partial x_i} := \lim_{h \rightarrow 0} \frac{u(x + he^i) - u(x)}{h}.$$

Definition 1.1.3. The gradient of u is the vector:

$$Du := \nabla u := (u_{x_1}, \dots, u_{x_n}).$$

Definition 1.1.4. The Hessian matrix of u is the 2 dimensional array:

$$D^2u = \begin{pmatrix} u_{x_1 x_1} & \dots & u_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ u_{x_n x_1} & \dots & u_{x_n x_n} \end{pmatrix}$$

Definition 1.1.5. The Laplacian of u is:

$$\Delta u := \nabla^2 u := \text{Tr}(D^2u) = \sum_{i=1}^n u_{x_i x_i}.$$

Definition 1.1.6.

- (a) A vector of the form $\alpha := (\alpha_1, \dots, \alpha_n)$, where $\alpha_i \in \mathbf{Z}_{\geq 0}$, is called a multiindex of order $|\alpha|$, where:

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

- (b) Given a multiindex α , we define:

$$D^\alpha u(x) := \frac{\partial^{|\alpha|} u(x)}{\partial^{\alpha_1} x_1 \dots \partial^{\alpha_n} x_n}.$$

Definition 1.1.7. A *partial differential equation* (PDE for short) is an equation involving an unknown function u along with one or more of its partial derivatives. A partial differential equation is linear if we can write it as:

$$\sum_{|a| \leq K} a_\alpha(x) D^\alpha u(x) = f(x).$$

If $f(x) = 0$, then this PDE is called homogenous.

§ 1.2. Multivariable Calculus

Definition 1.2.1. Let $U \subset \mathbb{R}^n$ and $f \in L^1(\mathbb{R}, \mu)$. The average of f over U is:

$$\oint f d\mu := \frac{1}{\mu(U)} \int_U f d\mu.$$

Theorem 1.2.1 (Fubini). If $u \in L^1(U)$, then:

$$\int_U u dx = \underbrace{\int \dots \int}_{n\text{-times}} u(x_1, \dots, x_n) dx_1 \dots dx_n.$$

Definition 1.2.2. The boundary of U is denoted ∂U .

Definition 1.2.3. We say the boundary ∂U is $\underline{C}^k$ if in a neighborhood of any point on the boundary, the boundary is the graph of a C^k function (possibly after the coordinate system has been rotated).

Definition 1.2.4. If ∂U is C^1 , then along ∂U is defined the outward pointing unit normal vector field, which we denote by:

$$\vec{v} = (v^1, \dots, v^n).$$

Definition 1.2.5. Let $u \in C^1(\overline{U})$. The outward normal derivative of u is:

$$\frac{\partial u}{\partial v} := Du \cdot \vec{v}.$$

Definition 1.2.6. Let $\vec{u} = (u^1, \dots, u^n)$ be a vector field. The flux of u across ∂U is:

$$\int_{\partial U} \vec{u} \cdot \vec{v} dS.$$

Theorem 1.2.2 (Gauss-Green).

(a) Suppose $u \in C^1(\overline{U})$. For any $i = 1, \dots, n$, we have:

$$\int_U u_{x_i} dx = \int_{\partial U} u v^i dS$$

(b) (Divergence Theorem) Let $\vec{v} = (v^1, \dots, v^n)$, where $v^i \in C^1(\overline{U})$ for each $i = 1, \dots, n$. Then:

$$\int_U \operatorname{div} \vec{v} dx = \int_{\partial U} \vec{v} \cdot \vec{\nu} dS.$$

Theorem 1.2.3 (Integration by Parts). Let $u, v \in C^1(\overline{U})$ and let ∂U be C^1 . Then:

$$\int_U u_{x_i} v dx = - \int_U u v_{x_i} dx + \int_{\partial U} u v \nu^i dS.$$

Theorem 1.2.4 (Green's Formulas). Let $u, v \in C^1(\overline{U})$ and let ∂U be C^1 . Then:

$$\begin{aligned} (a) \quad & \int_U \Delta u dx = \int_{\partial U} \frac{\partial u}{\partial \nu} dS, \\ (b) \quad & \int_U Dv \cdot Du dx = - \int_U (\Delta v) u dx + \int_{\partial U} \frac{\partial u}{\partial \nu} u dS, \\ (c) \quad & \int_U u \Delta v - v \Delta u dx = \int_{\partial U} u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} dS. \end{aligned}$$

§ 1.3. Polar Coordinates in $\mathbf{R}^n$

Remark. In $\mathbf{R}^n$, for any $x \neq 0$, we can write:

$$x = |x| \cdot \frac{x}{|x|} := r\omega,$$

where $r \in (0, \infty)$ and $\omega \in S^{n-1} = \partial B(0, 1)$.

Theorem 1.3.1. There exists a measure $d\omega$ on S^{n-1} such that:

$$\int_{\mathbf{R}^n} u dx = \int_0^\infty \left(\int_{S^{n-1}} u(r\omega) r^{n-1} d\omega \right) dr.$$

In other words, $dx = r^{n-1} d\omega dr$.

Corollary 1.3.2. For any $x_0 \in \mathbf{R}^n$ and $a > 0$, we have:

$$\int_{B(x_0, a)} u dx = \int_{B(0, a)} u(x_0 + y) dy = \int_0^a \left(\int_{S^{n-1}} u(x_0 + r\omega) r^{n-1} d\omega \right) dr.$$

Corollary 1.3.3. For any $x_0 \in \mathbf{R}^n$ and $r > 0$, we have:

$$\frac{d}{dr} \int_{B(x_0, r)} u dx = \int_{\partial B(x_0, r)} u dS.$$

§ 1.4. Smoothing

Definition 1.4.1.

- (a) Let $U, V \subset \mathbf{R}^n$. We say U is compactly contained in V , and write $U \subset\subset V$, if $V \subset \overline{V} \subset U$ and $\overline{V}$ is compact.
- (b) If $f|_V \in L^p(V)$ for all $V \subset\subset U$, then we say f is locally integrable on U . The set of all such functions is denoted $L^p_{\text{loc}}(U)$.

Definition 1.4.2.

- (a) Let $x \in \mathbf{R}^n$. The standard mollifier is the function:

$$\eta(x) := \begin{cases} \frac{-C}{1-|x|^2}, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases},$$

where C is chosen so that $\int_{\mathbf{R}^n} \eta(x) dx = 1$.

- (b) For all $\epsilon > 0$, define $\eta_\epsilon(x) := \frac{1}{\epsilon^n} \eta\left(\frac{x}{\epsilon}\right)$.

Remark. The standard mollifier is a radial function.

Proposition 1.4.1.

- (a) $\eta_\epsilon \in C_c^\infty(\mathbf{R}^n)$;
- (b) $\text{supp } \eta_\epsilon = B(0, \epsilon)$;
- (c) $\int_{\mathbf{R}^n} \eta_\epsilon(x) dx = 1$.

Theorem 1.4.2. Let $U \subset \mathbf{R}^n$ be open and $f \in L^1_{\text{loc}}(U)$. Define $U_\epsilon := \{y \in U : \epsilon < \text{dist}(y, \partial U)\}$ and $f^\epsilon(x) := (f * \eta_\epsilon)(x)$. Then:

- (a) $f^\epsilon \in C^\infty(U_\epsilon)$;
- (b) If $f \in C(U)$, then $f^\epsilon \rightarrow f$ uniformly on compact subsets of U .

Chapter 2

Classical Linear PDEs

§ 2.1. The Transport Equation

Definition 2.1.1. The Transport Equation is:

$$\left\{ u_t + b \cdot Du = 0, \quad \text{in } \mathbf{R}^n \times (0, \infty), \right.$$

where b is a fixed vector in $\mathbf{R}^n$ and $u : \mathbf{R}^n \times (0, \infty) \rightarrow \mathbf{R}$ is the unknown function.

Theorem 2.1.1. Let $g \in C^1(\mathbf{R}^n)$. Then $u(x, t) := g(x - tb)$ is a solution to the following initial value problem:

$$\left\{ \begin{array}{l} u_t + Du \cdot b = 0, \quad \text{in } \mathbf{R}^n \times (0, \infty) \\ u(x, 0) = g(x), \quad \text{on } \mathbf{R}^n \times \{0\} \end{array} \right.$$

Proof. As b is an arbitrary vector in $\mathbf{R}^n$, the trick is to parametrize the line through (x, t) with direction $(b, 1)$. Define $z(s) := u(x + sb, t + s)$. Then:

$$\begin{aligned} \dot{z}(s) &= \frac{d}{ds} u(x + sb, t + s) \\ &= \left(\sum_{j=1}^n u_{x_j}(x + sb, t + s) b_j \right) + u_t \\ &= Du \cdot b + u_t \\ &= 0. \end{aligned}$$

Since z is constant, we have for all $(x, t) \in \mathbf{R}^n \times (0, \infty)$:

$$\begin{aligned} u(x, t) &= z(0) \\ &= z(-t) \\ &= u(x - tb, 0) \\ &= g(x - tb). \end{aligned} \quad \blacksquare$$

Theorem 2.1.2. Let $g \in C^1(\mathbf{R}^n)$. Then $u(x, t) := g(x - tb) + \int_0^t f(x + (s+t)b, s) ds$ is a solution to the following non-homogenous initial value problem:

$$\left\{ \begin{array}{l} u_t + Du \cdot b = f, \quad \text{in } \mathbf{R}^n \times (0, \infty) \\ u(x, 0) = g(x), \quad \text{on } \mathbf{R}^n \times \{0\} \end{array} \right.$$

Proof. Similarly define $z(s) := u(x + sb, s + t)$. Then $\dot{z}(s) = Du \cdot b = f(x + sb, s + t)$. By the Fundamental Theorem of Calculus:

$$\begin{aligned} u(x, t) &= z(0) \\ &= z(-t) + \int_{-t}^0 f(x + sb, s + t) ds \\ &= g(x - tb) + \int_{-t}^0 f(x + sb, s + t) ds. \end{aligned}$$

Making the substitution $s \mapsto t + s$, gives the desired result. ■

§ 2.2. Laplace's Equation

Definition 2.2.1. Laplace's equation is:

$$\Delta u = 0.$$

A C^2 function satisfying Laplace's equation is called a harmonic function.

I. The Fundamental Solution

Proposition 2.2.1. *Laplace's equation is rotation invariant. Specifically, let $A = (a_{ij})$ denote an $n \times n$ orthogonal matrix. For $x \in \mathbb{R}^n$, let $y = Ax$; that is:*

$$Ax = y = (y_1, \dots, y_n), \quad y_i = \sum_{j=1}^n a_{ij} x_j$$

Assume $u \in C^2(\mathbb{R}^n)$ satisfies $\Delta u = 0$. Then define

$$v(x) = u(Ax) = u(y).$$

We have that $\Delta v = 0$.

Proof. Note that $\frac{\partial y_k}{\partial x_i} = a_{ki}$. Hence:

$$\begin{aligned} v_{x_i}(x) &= \frac{\partial}{\partial x_i} u(y) \\ &= \sum_{k=1}^n u_{y_k}(y) \frac{\partial y_k}{\partial x_i} \\ &= \sum_{k=1}^n u_{y_k}(y) a_{ki}. \end{aligned}$$

Taking a second partial derivative with respect to x_i gives:

$$\begin{aligned}
 v_{x_i x_i}(x) &= \frac{\partial}{\partial x_i} \sum_{k=1}^n u_{x_k}(y) a_{ki} \\
 &= \sum_{k=1}^n \sum_{j=1}^n u_{x_k x_j}(y) \frac{\partial y_j}{\partial x_i} a_{ki} \\
 &= \sum_{j=1}^n \sum_{k=1}^n a_{ji} a_{ki} u_{x_j x_k}.
 \end{aligned}$$

Summing from $i = 1, \dots, n$ finally yields:

$$\begin{aligned}
 \Delta v(x) &= \sum_{i=1}^n v_{x_i x_i}(x) \\
 &= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n a_{ji} a_{ki} u_{x_j x_k}(y) \\
 &= \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^n a_{ji} a_{ki} \right) u_{x_j x_k}(y) \\
 &= \sum_{j=1}^n \sum_{k=1}^n \delta_{jk} u_{x_j x_k}(y) \\
 &= \sum_{k=1}^n u_{x_k x_k} \\
 &= \Delta u(y) \\
 &= 0
 \end{aligned}$$

■

Example 2.2.1. Since solutions to Laplace's equation are rotation invariant, we are interested in finding *radial solutions* to the equation; i.e., functions of $r = |x|$. Let u be a solution of Laplace's equation and suppose further we have $u(x) = v(r)$. Then $u_{x_i} = v'(r)r_{x_i}$ and $u_{x_i x_i} = v''(r)r_{x_i}^2 + v'(r)r_{x_i x_i}$. Using the fact that:

$$\begin{aligned}
 \frac{\partial}{\partial x_i} r^2 &= \frac{\partial}{\partial x_i} \sum_{i=1}^n x_i^2 \\
 &= 2r r_{x_i},
 \end{aligned}$$

we get that $r_{x_i} = \frac{x_i}{r}$. A similar calculation yields $r_{x_i x_i} = \frac{1}{r} - \frac{x_i^2}{r^3}$. Now since u is a solution to Laplace's equation, we have that $\Delta u = 0$. We can write this as:

$$\sum_{i=1}^n u_{x_i x_i} = 0.$$

Substituting for our earlier calculation gives:

$$\sum_{i=1}^n v''(r) \frac{x_i^2}{r^2} + v'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right) = 0.$$

Hence:

$$v''(r) + v'(r) \frac{(n-1)}{r} = 0.$$

For $n = 2$, the solution to the above ODE is $v(r) = c_1 \log r + c_2$. For $n > 2$, the solution to the above ODE is $v(r) = c_1 \frac{1}{r^{n-1}} + c_2$.

Definition 2.2.2. The fundamental solution to Laplace's equation is:

$$\Phi(x) = \begin{cases} c_2 \log \frac{1}{|x|}, & n = 2 \\ c_n |x|^{2-n}, & n > 2, \end{cases}$$

where $c_2 = \frac{1}{2\pi}$ and $c_n = \frac{1}{n(n-2)\alpha(n)}$, where $\alpha(n)$ denotes the volume of $B(0, 1)$ in $\mathbf{R}^n$.

Remark. The fundamental solution to Laplace's equation admits a singularity at $x = 0$.

Remark. If u and v are both harmonic, then $au + bv$ is harmonic for any $a, b \in \mathbf{R}$ since the Laplacian is a linear differential operator. Moreover, if u is harmonic then $u(x - x_0)$ is also harmonic when treated as a variable with respect to x . These reasons combined lead to the following conclusion: if $y^1, \dots, y^k \in \mathbf{R}^n$ and $f_1, \dots, f_k \in \mathbf{R}$, the function:

$$\sum_{i=1}^k f_i \Phi(x - y^i)$$

is also harmonic! A natural question to ask now is whether:

$$\int_{\mathbf{R}^n} f(y) \Phi(x - y) dy$$

is harmonic.

Theorem 2.2.2. Let $f \in C_c^2(\mathbf{R})$. Define $u(x) = \int_{\mathbf{R}^n} f(y) \Phi(x - y) dy$. Then $u \in C^2(\mathbf{R}^n)$ and $-\Delta u = f$.

Proof. Taking the difference quotient of $u(x)$ gives the following equality:

$$\frac{u(x + he^i) - u(x)}{h} = \int_{\mathbf{R}^n} \frac{f(x + he^i - y) - f(x - y)}{h} \Phi(y) dy.$$

Applying the Dominated Convergence Theorem gives:

$$u_{x_i} = \int_{\mathbf{R}^n} f_{x_i}(x - y) \Phi(y) dy.$$

Taking the difference quotient of both sides and applying the Dominated Convergence Theorem again yields:

$$u_{x_i x_i} = \int_{\mathbf{R}^n} f_{x_i x_i}(x - y) \Phi(y) dy.$$

Thus:

$$\begin{aligned}\Delta u &= \int_{\mathbf{R}^n} \Delta_x f(x-y) \Phi(y) dy \\ &= \int_{\mathbf{R}^n} (-1)^2 \Delta_y f(x-y) \Phi(y) dy.\end{aligned}$$

Since f is a function with compact support, let $U := B^o(0, R)$ be large enough such that $x \in U$ and $\text{supp } f(x-y) \subset U$. Our integral becomes:

$$\begin{aligned}\Delta u &= \int_{\mathbf{R}^n} \Delta_y f(x-y) \Phi(y) dy \\ &= \int_U \Delta_y f(x-y) \Phi(y) dy.\end{aligned}$$

Recall that our fundamental solution Φ admits a discontinuity at the origin. For an arbitrary $\epsilon > 0$, we have:

$$\Delta u = \int_{B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy + \int_{U \setminus B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy$$

Since $\Delta_y f(x-y) \Phi(y)$ does not admit a discontinuity on $U \setminus B(0, \epsilon)$, we can apply Green's identities. Taking advantage of the fact that $\Delta_y \Phi(y) = 0$, we have:

$$\begin{aligned}\int_{U \setminus B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy &= \int_{U \setminus B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) - f(x-y) \Delta_y \Phi(y) dy \\ &= \int_{\partial(U \setminus B(0, \epsilon))} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) - f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y).\end{aligned}$$

Integrating on $\partial(U \setminus B(0, \epsilon))$ is equivalent to:

$$\int_{\partial U} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) - f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y) - \int_{\partial B(0, \epsilon)} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) - f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y).$$

The outward normal derivative of f is zero, and since f is of compact support it is zero along the boundary of U . The left-most integral is equal to zero, resulting in:

$$\Delta u = \underbrace{\int_{B(0, \epsilon)} \Delta_y f(x-y) \Phi(y) dy}_{:= I_\epsilon} - \underbrace{\int_{\partial B(0, \epsilon)} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) dS(y)}_{:= K_\epsilon} + \underbrace{\int_{\partial B(0, \epsilon)} f(x-y) \frac{\partial \Phi}{\partial \nu}(y) dS(y)}_{:= L_\epsilon}$$

We will individually examine I_ϵ , K_ϵ , and L_ϵ as $\epsilon \rightarrow \infty$. We first start with I_ϵ . Observe that:

$$\begin{aligned}
 |I_\epsilon| &\leq \int_{B(0,\epsilon)} |\Delta_y f(x-y)| \Phi(y) dy \\
 &\leq C \int_{B(0,\epsilon)} \Phi(y) dy \\
 &\leq C \int_{B(0,\epsilon)} |x|^{2-n} dy \\
 &= C \int_0^\epsilon \int_{S^{n-1}} r^{2-n} r^{n-1} dr d\omega \\
 &= C \int_0^\epsilon r \left(\int_{S^{n-1}} d\omega \right) dr \\
 &\leq C \int_0^\epsilon r dr \\
 &\leq C\epsilon^2.
 \end{aligned}$$

Thus $I_\epsilon \rightarrow 0$. We next examine K_ϵ . Observe that:

$$\begin{aligned}
 |K_\epsilon| &\leq \int_{\partial B(0,\epsilon)} \left| \frac{\partial f}{\partial \nu}(x-y) \right| \Phi(y) dy \\
 &\leq C \int_{\partial B(0,\epsilon)} \Phi(y) dS(y) \\
 &\leq C \int_{\partial B(0,\epsilon)} |y|^{2-n} dS(y) \\
 &= C\epsilon^{2-n} \int_{\partial B(0,\epsilon)} dS(y) \\
 &\leq C\epsilon^{2-n} \epsilon^{n-1} \\
 &= C\epsilon.
 \end{aligned}$$

Thus $K_\epsilon \rightarrow 0$. The last and most difficult integral to examine is L_ϵ . We must first compute $\frac{\partial \Phi}{\partial \nu}$. Computing a partial of Φ yields:

$$\begin{aligned}
 \frac{\partial}{\partial y_i} \Phi(y) &= c_n \frac{\partial}{\partial y_i} |y|^{2-n} \\
 &= c_n (2-n) |y|^{1-n} \frac{\partial}{\partial y_i} |y| \\
 &= -c_n (n-2) |y|^{1-n} \frac{1}{2} \left(\sum_{j=1}^n y_j^2 \right)^{-\frac{1}{2}} 2y_i \\
 &= -c_n (n-2) |y|^{1-n} \frac{y_i}{|y|} \\
 &= -c_n (n-2) |y|^{-n} y_i.
 \end{aligned}$$

Hence $D\Phi(y) = -c_n(n-2)\frac{1}{2}|y|^{2-n}y$. Since $y \in \partial B(0, \epsilon)$, it already points in the direction as the outward normal; i.e., $v = \frac{y}{|y|}$. With this:

$$\begin{aligned} \frac{\partial \Phi}{\partial v}(y) &= D\Phi(y) \cdot v \\ &= -c_n(n-2)|y|^{-n}y \cdot \frac{y}{|y|} \\ &= -c_n(n-2)|y|^{-n-1}(y \cdot y) \\ &= -c_n(n-2)|y|^{-n-1}|y|^2 \\ &= -c_n(n-2)|y|^{1-n}. \end{aligned}$$

But since we're integrating on $\partial B(0, \epsilon)$, we have $|y| = \epsilon$. The integral L_ϵ becomes:

$$\begin{aligned} L_\epsilon &= -c_n(n-2)\epsilon^{1-n} \int_{\partial B(0, \epsilon)} f(x-y) dS(y) \\ &= -c_n(n-2)\epsilon^{1-n} \frac{n\alpha(n)\epsilon^{n-1}}{n\alpha(n)\epsilon^{n-1}} \int_{\partial B(0, \epsilon)} f(x-y) dS(y) \\ &= -c_n(n-2)\epsilon^{1-n} n\alpha(n)\epsilon^{n-1} \oint_{\partial B(0, \epsilon)} f(x-y) dS(y) \\ &= - \oint_{\partial B(0, \epsilon)} f(x-y) dS(y). \end{aligned}$$

Thus $L_\epsilon \rightarrow -f(x)$ as $\epsilon \rightarrow 0$, establishing the theorem. ■

Definition 2.2.3. The equation:

$$-\Delta u = f$$

arising from Theorem 2.2.2 is called Poisson's equation.

II. Consequences of the Mean Value Property

Theorem 2.2.3 (Mean-Value Property). *Let $U \subset \mathbf{R}^n$ and $x \in U$. Suppose $u \in C^2(U)$ and $\Delta u = 0$ in U . For each $0 < r < \text{dist}(x, \partial U)$, we have:*

$$u(x) = \int_{B(x, r)} u(y) dy = \oint_{\partial B(x, r)} u dS.$$

Proof. Let $x \in U$. For each $0 < r < \text{dist}(x, \partial U)$, define:

$$\begin{aligned} \rho(r) &:= \int_{\partial B(x,r)} u \, dS \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} u \, dS \\ &= \frac{1}{n\alpha(n)r^{n-1}} \int_{S^{n-1}} u(x+r\omega)r^{n-1} \, d\omega \\ &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} u(x+r\omega) \, d\omega \end{aligned}$$

Taking the derivative of ρ with respect to r gives:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} \frac{d}{dr} u(x+r\omega) \, d\omega \\ &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(x+r\omega) \cdot \omega \, d\omega. \end{aligned}$$

Note that ω is precisely the outward normal on S^{n-1} . Making the substitution, $y = x + r\omega$ we have that $dy = r^{n-1} d\omega$ and $\omega = \frac{y-x}{r}$. Hence:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)} \int_{S^{n-1}} Du(x+r\omega) \cdot \omega \, d\omega \\ &= \frac{1}{n\alpha(n)r^{1-n}} \int_{\partial B(0,1)} Du(y) \cdot \frac{y-x}{r} \, dy. \end{aligned}$$

The final integrand is by definition the outward normal derivative of $u(y)$. Green's identity yields:

$$\begin{aligned} \rho'(r) &= \frac{1}{n\alpha(n)r^{1-n}} \int_{B(0,1)} \Delta u(y) \, dy \\ &= 0 \end{aligned}$$

Thus $\rho(r)$ is constant. With this fact, notice that.

$$\begin{aligned} \rho(r) &= \lim_{r \rightarrow 0^+} \rho(r) \\ &= u(x). \end{aligned}$$

This establishes $u(x) = \oint_{\partial B(x,r)} u \, dS$. Finally:

$$\begin{aligned}
 \oint_{B(x,r)} u(y) \, dy &= \frac{1}{\alpha(n)r^n} \int_{B(x,r)} u(y) \, dy \\
 &= \frac{1}{\alpha(n)r^n} \int_0^r \int_{S^{n-1}} u(x + \omega t) t^{n-1} \, d\omega \, dt \\
 &= \frac{1}{\alpha(n)r^n} \int_0^r t^{n-1} \int_{S^{n-1}} u(x + \omega t) \, d\omega \, dt \\
 &= \frac{1}{\alpha(n)r^n} \int_0^r n\alpha(n)t^{n-1} u(x) \, dt \\
 &= u(x). \quad \blacksquare
 \end{aligned}$$

Theorem 2.2.4. *If $u \in C^2(U)$ satisfies $u(x) = \oint_{\partial B(x,r)} u \, dS$ for every $B(x,r) \subset U$, then $\Delta u = 0$.*

Proof. Suppose towards contradiction $\Delta u(x_0) > 0$ for some $x_0 \in U$. Then there exists some $\epsilon > 0$ such that $\Delta u(y) > 0$ for all $y \in B(x_0, \epsilon)$. For $0 < r < \epsilon$, define $\rho(r) := \oint_{\partial B(x_0,r)} u(y) \, dy$. By the same calculations made in Theorem 2.2.3, we have $\rho'(r) > 0$. But since $\rho(r) = u(x_0)$ by hypothesis, we also have $\rho'(r) = 0$. This is a contradiction, hence $\Delta u = 0$. $\blacksquare$

Theorem 2.2.5. *Let $U \subset \mathbb{R}^n$. Let $u \in C(U)$. If u satisfies:*

$$u(x) = \oint_{\partial B(x,r)} u \, dS = \oint_{B(x,r)} u(y) \, dy,$$

for any $B(x,r) \subset U$, then $u \in C^\infty(U)$.

Proof. Let $x \in U$. Choose $0 < \epsilon < \text{dist}(x, \partial U)$. Then:

$$\begin{aligned}
 u^\epsilon(x) &= (u * \eta_\epsilon)(x) \\
 &= \int_{\mathbb{R}^n} u(x-y) \eta_\epsilon(y) \, dy.
 \end{aligned}$$

Since η_ϵ is zero outside of $B(0, \epsilon)$, we have:

$$\begin{aligned}
 u^\epsilon(x) &= \int_{B(0,\epsilon)} u(x-y) \eta_\epsilon(y) \, dy \\
 &= \int_0^\epsilon \int_{S^{n-1}} u(x-r\omega) \eta_\epsilon(r\omega) r^{n-1} \, d\omega \, dr.
 \end{aligned}$$

Since η_ϵ is a radial function, it does not depend on ω , allowing us to rearrange the above integral as follows:

$$u^\epsilon(x) = \int_0^\epsilon \eta_\epsilon(r\omega) r^{n-1} \left(\int_{S^{n-1}} u(x-r\omega) \, d\omega \right) dr.$$

By hypothesis, $u(x) = \int_{\partial B(x,r)} u dS$ for any $B(x,r) \subset U$. Recalling that the surface area of the unit n -sphere is $n\alpha(n)$, we can rewrite the inner integral as:

$$\begin{aligned} u^\epsilon(x) &= \int_0^\epsilon \eta_\epsilon(r\omega) r^{n-1} (n\alpha(n)u(x)) dr \\ &= u(x) \int_0^\epsilon \eta_\epsilon(r\omega) r^{n-1} (n\alpha(n)) dr. \end{aligned}$$

But $n\alpha(n) = \int_{S^{n-1}} d\omega$. Substituting and rearranging gives:

$$\begin{aligned} u^\epsilon(x) &= u(x) \int_0^\epsilon \int_{S^{n-1}} \eta_\epsilon(r\omega) r^{n-1} d\omega dr \\ &= u(x) \int_{B(0,\epsilon)} \eta_\epsilon(y) dy \\ &= u(x). \end{aligned}$$

Thus $u^\epsilon(x) = u(x)$ on U_ϵ . By Theorem ??, this establishes $u \in C^\infty(U_\epsilon)$. To conclude $u \in C^\infty(U)$, for any compact set $K \subset U$ and $\epsilon < \text{dist}(K, \partial U)$, we have $K \subset U_\epsilon$, hence $u = u^\epsilon$ on K . Since K was arbitrary, we are done. ■

Corollary 2.2.6. *Let $U \subset \mathbf{R}^n$. If u satisfies $\Delta u = 0$ in U , then $u \in C^\infty(U)$.*

Theorem 2.2.7 (Maximum Principle). *Let $U \subset \mathbf{R}^n$ and $u \in C^2(U) \cap C(\overline{U})$. Assume $\Delta u = 0$ in U . Then:*

- (a) (Weak Maximum Principle) $\max_{\overline{U}} u = \max_{\partial U} u$;
- (b) (Strong Maximum Principle) If U is connected, and there exists $x_0 \in U$ such that $u(x_0) = \max_{\overline{U}} u$, then u is constant on U .

Proof. (a) As $\partial U \subseteq \overline{U}$, we immediately have $\max_{\partial U} u \leq \max_{\overline{U}} u$. For the other direction of inequality, let $\epsilon > 0$. Consider the function $u + \epsilon|x|^2$. Then:

$$\begin{aligned} \Delta(u + \epsilon|x|^2) &= \Delta u + \epsilon \Delta|x|^2 \\ &= \epsilon \sum_{j=1}^n \frac{\partial}{\partial x_j} \left(\frac{\partial}{\partial x_j} |x|^2 \right) \\ &= \epsilon \sum_{j=1}^n \frac{\partial}{\partial x_j} 2x_j \\ &= \epsilon \sum_{j=1}^n 2 \\ &= 2n\epsilon \\ &> 0. \end{aligned}$$

If $u + \epsilon|x|^2$ were to attain a local maximum at an interior point, then $\frac{\partial^2}{\partial x_j^2} (u + \epsilon|x|^2) \leq 0$ for all $j = 1, \dots, n$. This immediately implies that $\Delta(u + \epsilon|x|^2) \leq 0$, contradicting the above result.

Since the local maximum of $u + \epsilon|x|^2$ must lie on its boundary, we have:

$$\begin{aligned}\max_{\overline{U}} u &\leq \max_{\overline{U}} (u + \epsilon|x|^2) \\ &= \max_{\partial U} (u + \epsilon|x|^2)\end{aligned}$$

Taking $\epsilon \rightarrow 0$ gives $\max_{\overline{U}} u \leq \max_{\partial U} u$, establishing the claim.

(b) Assume $u(x_0) = \max_{\overline{U}} u := M$. Let $B(x_0, r) \subset U$. By the Mean Value Property:

$$\begin{aligned}M &= u(x_0) \\ &= \oint_{B(x_0, r)} u(x) dx \\ &= \frac{1}{\alpha(n)r^n} \int_{B(x_0, r)} u(x) dx \\ &\leq \frac{M}{\alpha(n)r^n} \int_{B(x_0, r)} dx \\ &= M.\end{aligned}$$

If the maximum of $B(x_0, r)$ is $u(x_0)$, and if $\oint_{B(x_0, r)} u(x) dx = M$, it must be the case that $u(x) = M$ on all of $B(x_0, r)$. Since u is locally constant on a connected space, it is globally constant; i.e., $u(x) = M$ for all $x \in U$. ■

Remark. Note that the strong maximum principle implies the weak maximum principle.

Remark. A similar theorem can be stated in terms of local minima of functions, and the proof follows by consider $-u(x)$ and applying the maximum principle.

Corollary 2.2.8. Let $y \in C(\partial U)$ and $f \in C(U)$. Solutions to the boundary value problem:

$$\begin{cases} \Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases}$$

are unique.

Proof. Let u and v be solutions to the above boundary value problem. Since the Laplacian is linear, this induces the following boundary value problem:

$$\begin{cases} \Delta(u - v) = 0, & \text{in } U \\ u - v = 0, & \text{on } \partial U. \end{cases}$$

Since $u - v$ is harmonic in U , Theorem 2.2.7 and Theorem ?? assert that $\max_U(u - v) = \min_U(u - v)$. It must be the case that $u - v = 0$ everywhere on U , hence $u = v$. ■

Theorem 2.2.9 (Interior Derivate Estimate). Assume $\Delta u = 0$ in U . Then:

$$|Du| \leq \frac{C}{r^{n+1}} \|u\|_{L^1(B(x_0, r))}$$

for each $B(x_0, r) \subset U$.

Proof. First, note that if u is a harmonic function then u_{x_i} is also harmonic (Clairaut's Theorem). As such, the function u_{x_i} admits the Mean-Value Property:

$$\begin{aligned}
 |u_{x_i}(x_0)| &= \left| \oint_{B(x_0, \frac{r}{2})} u_{x_i} dx \right| \\
 &= \frac{2^n}{\alpha(n)r^n} \left| \int_{B(x_0, \frac{r}{2})} u_{x_i} dx \right| \\
 &= \frac{2^n}{\alpha(n)r^n} \left| \int_{\partial B(x_0, \frac{r}{2})} u v^i dS \right| \\
 &\leq \frac{2^n}{\alpha(n)r^n} \sup \left\{ |u(x)| : x \in \partial B\left(x_0, \frac{r}{2}\right) \right\} \int_{\partial B(x_0, \frac{r}{2})} dS \\
 &= \frac{2^n}{\alpha(n)r^n} \|u\|_{L^\infty(\partial B(x_0, \frac{r}{2}))} n\alpha(n) \frac{r^{n-1}}{2^{n-1}} \\
 &= \frac{2n}{r} \|u\|_{L^\infty(\partial B(x_0, \frac{r}{2}))}
 \end{aligned}$$

Note that for any $x \in \partial B(x_0, \frac{r}{2})$, we have $B(x, \frac{r}{2}) \subset B(x_0, r)$. Using this fact we can estimate $|u(x)|$ as follows:

$$\begin{aligned}
 |u(x)| &= \left| \oint_{B(x, \frac{r}{2})} u(y) dy \right| \\
 &= \frac{2^n}{\alpha(n)r^n} \int_{B(x, \frac{r}{2})} |u(y)| dy \\
 &= \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x, \frac{r}{2}))} \\
 &\leq \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))}.
 \end{aligned}$$

Taking the supremum over all $x \in \partial B(x_0, \frac{r}{2})$ gives:

$$\|u\|_{L^\infty(\partial B(x_0, \frac{r}{2}))} \leq \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))}.$$

Combining this with our first string of inequalities results in:

$$\begin{aligned}
 |u_{x_i}(x_0)| &\leq \frac{2n}{r} \cdot \frac{2^n}{\alpha(n)r^n} \|u\|_{L^1(B(x_0, r))} \\
 &= \frac{n2^{n+1}}{\alpha(n)r^{n+1}} \|u\|_{L^1(B(x_0, r))}.
 \end{aligned}$$

Defining $C := \frac{n2^{n+1}}{\alpha(n)}$ and taking the gradient of $u_{x_i}(x_0)$ gives the desired result. ■

Theorem 2.2.10. Assume $\Delta u = 0$ in U . Then:

$$|D^\alpha u(x_0)| \leq \frac{C_k}{r^{n+|\alpha|}} \|u\|_{L^1(B(x_0, r))}$$

for each $B(x_0, r) \subset U$.

Proof. ■

Corollary 2.2.11 (Liouville). *Suppose $u : \mathbf{R}^n \rightarrow \mathbf{R}$ is harmonic. If u is bounded, then u is constant.*

Proof. Let $x_0 \in \mathbf{R}^n$ and $r > 0$. Assume $|u(x)| < M$. For each $i = 1, \dots, n$ we have:

$$\begin{aligned} |u_{x_i}(x_0)| &\leq \frac{C}{r^{n+1}} \|u\|_{L^1(B(x_0, r))} \\ &\leq \frac{C}{r^{n+1}} \int_{B(x_0, r)} M dx \\ &\leq \frac{CM}{r^{n+1}} \alpha(n) r^n \\ &= \frac{CM\alpha(n)}{r}. \end{aligned}$$

As we are assuming u is harmonic on all of $\mathbf{R}^n$, taking $r \rightarrow \infty$ does not fail any of the necessary assumptions made in Theorem 2.2.9. Since $u_{x_i}(x_0) = 0$, we conclude that u is constant. ■

III. Green's Function, Poisson's Formula, and Analyticity

Example 2.2.2. Let $U \subset \mathbf{R}^n$ be open, bounded, and ∂U is C^1 . Suppose $u \in C^2(U) \cap C^1(\overline{U})$. Can we write a formula for u in terms of its boundary values? Fix $x \in U$ and choose $\epsilon > 0$ such that $B(x, \epsilon) \subset U$. As a trick, consider Green's Formula III applied to the following integral:

$$\begin{aligned} &\int_{U \setminus B(x, \epsilon)} u(y) \Delta \Phi(y-x) - \Phi(y-x) \Delta u(y) dy \\ &= \int_{\partial(U \setminus B(x, \epsilon))} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x) \frac{\partial u}{\partial \nu}(y) dS(y) \\ &= \int_{\partial U} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x) \frac{\partial u}{\partial \nu}(y) dS(y) - \underbrace{\int_{\partial B(x, \epsilon)} u(y) \frac{\partial \Phi}{\partial \nu}(y-x)}_{(*)} - \underbrace{\Phi(y-x) \frac{\partial u}{\partial \nu}(y) dS(y)}_{(**)}. \end{aligned}$$

First, note that $\Delta \Phi(x) = 0$, so our original integral can be simplified. Moreover, the derivations of $|K_\epsilon|$ and $|L_\epsilon|$ as $\epsilon \rightarrow 0$ from Theorem ?? can be replicated on $(**)$ and $(*)$ respectively. As such, we have:

$$\begin{aligned} &\left| \int_{\partial B(x, \epsilon)} \Phi(y-x) \frac{\partial u}{\partial \nu}(y) dS(y) \right| \rightarrow 0 \\ &\quad - \int_{\partial B(x, \epsilon)} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) \rightarrow u(x). \end{aligned}$$

All together, sending $\epsilon \rightarrow 0$ gives:

$$u(x) = - \int_U \Phi(y-x) \Delta u(y) dy - \int_{\partial U} u(y) \frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x) \frac{\partial u}{\partial \nu}(y) dS(y)$$

The first term of this integral is sticking around unless we assume u to be harmonic. The term $u(y) \frac{\partial \Phi}{\partial \nu}(y-x)$ is exactly what we were hoping for—it requires us to know information about u along its boundary. The term $\Phi(x-y) \frac{\partial u}{\partial \nu}$ is problematic however. With the given information, we are unable to compute $\frac{\partial u}{\partial \nu}(y)$. Our goal is to construct a *corrector function* which cancels with the unknown term.

Definition 2.2.4. Let $U \subset \mathbf{R}^n$. For $x, y \in U$ with $x \neq y$, Green's function for the region U is:

$$G(x, y) := \Phi(y-x) - \phi^x(y),$$

where $\phi^x(y)$ is a corrector function which solves the following boundary-value problem:

$$\begin{cases} \Delta \phi^x = 0, & \text{in } U \\ \phi^x = \Phi(y-x), & \text{on } \partial U \end{cases}.$$

Theorem 2.2.12. If $u \in C^2(\overline{U})$ solves the boundary-value problem:

$$\begin{cases} -\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases},$$

then:

$$u(x) = - \int_{\partial U} g(y) \frac{\partial G}{\partial \nu}(x, y) dS(y) + \int_U f(y) G(x, y) dy$$

Proof. ■

Theorem 2.2.13. Suppose u solves the boundary-value problem:

$$\begin{cases} \Delta u = 0, & \text{in } B^o(0, 1) \\ u = g, & \text{on } \partial B(0, 1) \end{cases}.$$

Then:

$$u(x) = \frac{1-|x|^2}{n\alpha(n)} \int_{\partial B(0,1)} \frac{g(y)}{|x-y|^n} dS(y).$$

Proof. We must determine the corrector function:

$$\begin{cases} \Delta \phi^x = 0, & \text{in } B^o(0, 1) \\ \phi^x(y) = \Phi(y-x), & \text{on } \partial U \end{cases}.$$

Claim: $\phi^x(y) := \Phi(|x|(y-\tilde{x}))$ satisfies the above BVP, where $\tilde{x} := \frac{x}{|x|^2}$. Clearly $\phi^x(y)$ is harmonic for each $y \in B^o(0, 1)$. Now let $y \in \partial B(0, 1)$. ■

IV. Harnack's Inequality

Theorem 2.2.14 (Harnack's Inequality). *Let $U \subset \mathbf{R}^n$ be open and bounded. Assume $\Delta u = 0$ in U and $u(x) \geq 0$ for all $x \in U$. For any $V \subset\subset U$ which is open and connected, there exists a constant $C(U, V)$ depending only on U and V such that $\sup_V u \leq C \inf_V u$.*

Proof. Let $r = \frac{1}{4} \text{dist}(V, \partial U)$. Let $x, y \in V$ with $|x - y| < r$. Then $B(x, r) \subset B(y, 2r)$. By the Mean Value Principle:

$$\begin{aligned} u(x) &= \oint_{B(x, r)} u(z) dz \\ &= \frac{1}{\alpha(n)r^n} \int_{B(x, r)} u(z) dz \\ &\leq \frac{1}{\alpha(n)r^n} \int_{B(y, 2r)} u(z) dz \\ &= \frac{2^n}{\alpha(n)r^n 2^n} \int_{B(y, 2r)} u(z) dz \\ &= 2^n u(y). \end{aligned}$$

Let $\{B_i\}_{i=1}^N$ be an open cover of $\overline{V}$ with finitely many balls of radius $\frac{r}{2}$. Since V is connected, for any $x, y \in V$ there is a chain $B_{i_1}, \dots, B_{i_\ell}$ such that $x \in B_{i_1}$, $y \in B_{i_\ell}$, and $B_{i_j} \cap B_{i_{j+1}} \neq \emptyset$ for each $j = 1, \dots, \ell - 1$. Pick an arbitrary $y_j \in B_{i_j} \cap B_{i_{j+1}}$ ¹. Then by our previous calculation:

$$\begin{aligned} u(x) &\leq 2^n u(y_1) \\ &\leq 2^n 2^n u(y_2) \\ &\vdots \\ &\leq 2^{\ell n} u(y). \end{aligned}$$

As $x \in V$ was arbitrary, we have $\sup_V u \leq C u(y)$. As $y \in V$ was arbitrary, we have $\sup_V u \leq C \inf_V u$. ■

V. Energy Methods

Below is an alternative proof of uniqueness for the boundary value problem in Corollary 2.2.8 that does not rely on the Mean Value Property.

Lemma 2.2.15. *Let $U \subset \mathbf{R}^n$ be open, bounded, and ∂U is C^1 . Suppose $u \in C^2(\overline{U})$. If:*

$$\begin{cases} \Delta u = 0, & \text{in } U \\ u = 0, & \text{on } \partial U \end{cases},$$

then $u \equiv 0$.

¹We don't need to be concerned that the balls intersect only at a single point. As the open balls form a base for $\mathbf{R}^n$, we can find another open ball contained in each of their intersections.

Proof. We employ the following trick and then apply integration by parts:

$$\begin{aligned}
 0 &= - \int_U u \Delta u \, dx \\
 &= - \int_U \sum_{i=1}^n u u_{x_i x_i} \, dx \\
 &= - \sum_{i=1}^n \int_U u u_{x_i x_i} \, dx \\
 &= - \sum_{i=1}^n \left(- \int_U u_{x_i} u_{x_i} \, dx + \int_{\partial U} u u_{x_i} \nu^i \, dS \right) \\
 &= \int_U |Du|^2 \, dx.
 \end{aligned}$$

Hence Du is zero, meaning that u is constant. Since u attains its maximum and minimum on its boundary, it must be the case that $u \equiv 0$. ■

Theorem 2.2.16. Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1 . Suppose $u \in C^2(\overline{U})$ satisfies:

$$\begin{cases} -\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases}$$

Then u is unique.

Proof. If u and v are solutions to the above boundary value problem, then $u - v$ satisfies the necessary conditions in Lemma 2.2.15. ■

Theorem 2.2.17 (Dirichlet Principle). Let $U \subset \mathbb{R}^n$ be open, bounded, and ∂U is C^1 . Consider the boundary value problem:

$$\begin{cases} -\Delta u = f, & \text{in } U \\ u = g, & \text{on } \partial U \end{cases}$$

where $f \in C(U)$ and $g \in C(\partial U)$. Define:

$$\mathcal{A} := \{w \in C^2(\overline{U}) : w = g \text{ on } \partial U\}, \quad I[w] := \int_U \left(\frac{1}{2} |Dw|^2 - fw \right) dx$$

(a) If $u \in \mathcal{A}$ satisfies the above BVP, then u minimizes I .

(b) If $u \in \mathcal{A}$ minimizes I , then u satisfies the above BVP.

Proof. (a) Let $u \in C^2(\overline{U})$ satisfy the above BVP. Let $w \in \mathcal{A}$. Then:

$$\begin{aligned}
 I[w] &= I[u + (w - u)] \\
 &= \int_U \frac{1}{2} |D(u + (w - u))|^2 - f(u + (w - u)) dx \\
 &= \int_U \frac{1}{2} |Du + D(w - u)|^2 - fu - f(w - u) dx \\
 &= \int_U \frac{1}{2} (|Du|^2 + 2Du \cdot D(w - u) + |D(w - u)|^2) - fu - f(w - u) dx \\
 &= \int_U \frac{1}{2} |Du|^2 - fu dx + \int_U Du \cdot D(w - u) - f(w - u) dx + \int_U \frac{1}{2} |D(w - u)|^2 dx \\
 &= I[u] + \int_U Du \cdot D(w - u) dx - \int_U f(w - u) dx + \int_U \frac{1}{2} |D(w - u)|^2 dx.
 \end{aligned}$$

Applying Greens Formula II to $\int_U Du \cdot D(w - u) dx$ gives:

$$\int_U Du \cdot D(w - u) dx = - \int_U (w - u) \Delta u dx + \int_{\partial U} v \cdot D(w - u) u dS.$$

Note that $\Delta u = f$ in U and $w - u = 0$ on ∂U . Hence:

$$\int_U (w - u) \Delta u dx = - \int_U (w - u) f dx.$$

The equation for $I[w]$ now simplifies to:

$$\begin{aligned}
 I[w] &= I[u] + \int_U \frac{1}{2} |D(w - u)|^2 dx \\
 &\geq I[u].
 \end{aligned}$$

Thus u minimizes I .

(b) Suppose $u \in \mathcal{A}$ minimizes I . Fix $v \in C_c^\infty(U)$. Define $i(t) := I[u + tv]$. Then:

$$\begin{aligned}
 0 &= i'(0) \\
 &= \int_U (Du \cdot Dv - fv) dx \\
 &= \int_U (-v \Delta u - fv) dx \\
 &= \int_U (-\Delta u - f)v dx
 \end{aligned}$$

It must be the case that $-\Delta u = f$ in U . Hence u satisfies the above BVP. ■

§ 2.3. The Heat Equation

I. The Fourier Transform

Definition 2.3.1. The Fourier transform of a function $u \in L^1(\mathbf{R}^n)$ is defined by:

$$\hat{u}(y) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbf{R}^n} u(x) e^{-i(x \cdot y)} dx.$$

The inverse Fourier transform is given by:

$$\check{u}(y) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbf{R}^n} u(x) e^{i(x \cdot y)} dx.$$

Lemma 2.3.1. For $x \in \mathbf{R}^n$ and $t > 0$, define $\Phi(x, t) := \frac{1}{(4\pi t)^{\frac{n}{2}}} e^{-\frac{|x|^2}{4t}}$.

$$(1) \int_{\mathbf{R}^n} \Phi(x, t) dx = 1.$$

$$(2) \text{ If } w \in C(\mathbf{R}^n) \cap L^1(\mathbf{R}^n), \text{ then } \lim_{t \rightarrow 0^+} \int_{\mathbf{R}^n} \Phi(x, t) w(x) dx = w(0).$$

Proof. (1) Rewrite Φ as:

$$\Phi(x, t) := \frac{1}{(4\pi t)^{\frac{n}{2}}} e^{-\left|\frac{x}{\sqrt{4t}}\right|^2}.$$

Take $y := \frac{x}{\sqrt{4t}}$. Then $dy = \frac{dx}{(4t)^{\frac{n}{2}}}$. Fubini's Theorem will yield:

$$\begin{aligned} \int_{\mathbf{R}^n} \Phi(x, t) dx &= \frac{1}{(4\pi t)^{n/2}} \int_{\mathbf{R}^n} e^{-|y|^2} (4t)^{\frac{n}{2}} dy \\ &= \frac{1}{\pi^{\frac{n}{2}}} \int_{\mathbf{R}^n} e^{-|y|^2} dy \\ &= \frac{1}{\pi^{\frac{n}{2}}} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} e^{-\sum_{i=1}^n y_i^2} dy_1 \dots dy_n \\ &= \left(\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y_1^2} dy_1 \right) \dots \left(\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y_n^2} dy_n \right) \\ &= 1. \end{aligned}$$

(2) Fix $t > 0$. Making the same substitution as above gives:

$$\begin{aligned} \int_{\mathbf{R}^n} \Phi(x, t) w(x) dx &= \int_{\mathbf{R}^n} \frac{1}{(4\pi t)^{\frac{n}{2}}} e^{-\frac{|x|^2}{4t}} w(x) dx \\ &= \frac{1}{\pi^{\frac{n}{2}}} \int_{\mathbf{R}^n} e^{-|y|^2} w(\sqrt{4t}y) dy. \end{aligned}$$

Hence:

$$\begin{aligned} \lim_{t \rightarrow 0^+} \int_{\mathbf{R}^n} \Phi(x, t) w(x) dx &= \lim_{t \rightarrow 0^+} \int_{\mathbf{R}^n} e^{-|y|^2} w(\sqrt{4t}y) dy \\ &= \frac{1}{\pi^{\frac{n}{2}}} \int_{\mathbf{R}^n} e^{-|y|^2} \left(\lim_{t \rightarrow 0^+} w(\sqrt{4t}y) \right) dy \\ &= \frac{w(0)}{\pi^{\frac{n}{2}}} \int_{\mathbf{R}^n} e^{-|y|^2} dy \\ &= w(0). \end{aligned}$$

■

Lemma 2.3.2. *If $u, v \in L^1(\mathbf{R}^n)$, then $(u * v) \in L^1(\mathbf{R}^n)$.*

Proof. Let $w := u * v$. We have:

$$\begin{aligned}
 \int_{\mathbf{R}^n} |w(y)| dy &= \int_{\mathbf{R}^n} \left| \int_{\mathbf{R}^n} u(y-x)v(x) dx \right| dy \\
 &\leq \int_{\mathbf{R}^n} \int_{\mathbf{R}^n} |u(y-x)||v(x)| dx dy \\
 &\leq \int_{\mathbf{R}^n} \int_{\mathbf{R}^n} |u(y-x)||v(x)| dy dx \\
 &= \int_{\mathbf{R}^n} \left(\int_{\mathbf{R}^n} |u(y-x)| dy \right) |v(x)| dx \\
 &= \int_{\mathbf{R}^n} \|u\|_{L^1(\mathbf{R}^n)} |v(x)| dx \\
 &= \|u\|_{L^1(\mathbf{R}^n)} \|v\|_{L^1(\mathbf{R}^n)}.
 \end{aligned}$$

Thus $w \in L^1(\mathbf{R}^n)$. ■

Lemma 2.3.3. *If $u, v \in L^1(\mathbf{R}^n) \cap L^2(\mathbf{R}^n)$, then $(u * v) \in C(\mathbf{R}^n)$.*

Proof. Define $w := u * v$. Let $\epsilon > 0$. Find $\varphi \in C_c^\infty(\mathbf{R}^n)$ such that $\|\varphi - u\|_{L^2(\mathbf{R}^n)} < \epsilon$. Since φ is continuous, find $\delta > 0$ such that for any $x, y \in \mathbf{R}^n$ we have $|x - y| < \delta$ implies $|\varphi(x) - \varphi(y)| < \epsilon$. If $|x - y| < \delta$, then:

$$\begin{aligned}
 |w(x) - w(y)| &= \left| \int_{\mathbf{R}^n} u(x-z)v(z) dz - \int_{\mathbf{R}^n} u(y-z)v(z) dz \right| \\
 &\leq \int_{\mathbf{R}^n} |u(x-z) - u(y-z)|v(z) dz
 \end{aligned}$$

We can rewrite this final integral as:

$$\int_{\mathbf{R}^n} |u(x-z) - \varphi(x-z)||v(z)| dz + \int_{\mathbf{R}^n} |\varphi(x-z) - \varphi(y-z)||v(z)| dz + \int_{\mathbf{R}^n} |\varphi(y-z) - u(y-z)||v(z)| dz.$$

Note that if $|x - y| < \delta$, then $|(x-z) - (y-z)| < \delta$. Using this on the second integral, as well as applying Cauchy-Schwartz to the first and third integral gives:

$$\begin{aligned}
 |w(x) - w(y)| &\leq 2 \|u - \varphi\|_{L^2(\mathbf{R}^n)} \|v\|_{L^2(\mathbf{R}^n)} + \epsilon \|v\|_{L^2(\mathbf{R}^n)} \\
 &< 2\epsilon \|v\|_{L^2(\mathbf{R}^n)} + \epsilon \|v\|_{L^2(\mathbf{R}^n)}.
 \end{aligned}$$
■

Theorem 2.3.4 (Plancharel). *Let $u \in L^1(\mathbf{R}^n) \cap L^2(\mathbf{R}^n)$. Then $\hat{u}, \check{u} \in L^2(\mathbf{R}^n)$ and:*

$$\|\hat{u}\|_{L^2(\mathbf{R}^n)} = \|u\|_{L^2(\mathbf{R}^n)} = \|\check{u}\|_{L^2(\mathbf{R}^n)}.$$

Proof. We will first show that if $v, w \in L^1(\mathbf{R}^n)$, then $\hat{v}, \hat{w} \in L^\infty(\mathbf{R}^n)$. Indeed:

$$\begin{aligned} |\hat{v}(y)| &= \left| \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbf{R}^n} v(x) e^{-i(x \cdot y)} dx \right| \\ &\leq \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbf{R}^n} |v(x)| dx \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \|v\|_{L^1(\mathbf{R}^n)}. \end{aligned}$$

This is true for all $y \in \mathbf{R}^n$, hence $\|u\|_{L^\infty(\mathbf{R}^n)} < \infty$. The same argument applies for w . With this, we have that $\int_{\mathbf{R}^n} v(x) \hat{w}(x) dx$ exists and is equal to $\int_{\mathbf{R}^n} \hat{v}(y) w(y) dy$. Indeed:

$$\begin{aligned} \int_{\mathbf{R}^n} v(x) \hat{w}(x) dx &= \int_{\mathbf{R}^n} v(x) \left(\frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbf{R}^n} w(y) e^{-i(x \cdot y)} dy \right) dx \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbf{R}^n} \int_{\mathbf{R}^n} v(x) w(y) e^{-i(x \cdot y)} dx dy \end{aligned}$$

■

Definition 2.3.2. For $u \in L^2(\mathbf{R}^n)$, let $\{u_k\}_{k=1}^\infty$ be a Cauchy sequence in $L^1(\mathbf{R}^n) \cap L^2(\mathbf{R}^n)$ such that $u_k \rightarrow u$ in $L^2(\mathbf{R}^n)$. Then $\{\hat{u}_k\}_{k=1}^\infty$ is also Cauchy in $L^2(\mathbf{R}^n)$ and we define the Fourier transform of u to be $\hat{u} := \lim_{k \rightarrow \infty} \hat{u}_k$.

Theorem 2.3.5. Let $u, v \in L^2(\mathbf{R}^n)$.

- (1) $\int_{\mathbf{R}^n} u \bar{v} dx = \int_{\mathbf{R}^n} \hat{u} \bar{\hat{v}} dy$.
- (2) $(D^\alpha u)^\wedge = (iy)^\alpha \hat{u}$ for each multiindex α such that $D^\alpha u \in L^2(\mathbf{R}^n)$.
- (3) If $u, v \in L^1(\mathbf{R}^n) \cap L^2(\mathbf{R}^n)$, then $(u * v)^\wedge = (2\pi)^{\frac{n}{2}} \hat{u} \hat{v}$.
- (4) $u = (\hat{\hat{u}})^\vee$.

Proof.

■

Example 2.3.1. Let $u \in C^2(\mathbf{R})$. Consider the ODE:

$$u'' - u = f.$$

Applying the Fourier Transform to both sides gives:

$$\hat{u}''(y) - \hat{u}(y) = \hat{f}(y).$$

In general, note that:

$$\begin{aligned}
 \hat{u}'(y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u'(x) e^{-xy} dx \\
 &= \frac{1}{\sqrt{2\pi}i} \left(f(x) e^{-ixy} \Big|_{-\infty}^{\infty} + iy \int_{-\infty}^{\infty} u(x) e^{-xy} dx \right) \\
 &= iy \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x) e^{-ixy} dx \\
 &= iy \hat{u}(y)
 \end{aligned}$$

This means:

$$(iy)^2 \hat{u}(y) - \hat{u}(y) = \hat{f}(y).$$

Solving for $\hat{u}(y)$ yields:

$$\hat{u}(y) = \frac{\hat{f}(y)}{1 + y^2}.$$

Taking the inverse Fourier transform on both sides gives:

$$u(x) = \frac{1}{\sqrt{2\pi}} \left(f(x) * \left(\frac{1}{1 + x^2} \right)^{\vee} \right)$$

II. The Heat Equation

Chapter 3

Functional Analytic Methods for PDEs

§ 3.1. Weak Derivatives and Sobolev Spaces

I. Hölder spaces

Definition 3.1.1. Let $0 < \gamma \leq 1$. A function $u : U \rightarrow \mathbf{R}$ is Hölder continuous with exponent γ if:

$$|u(x) - u(y)| \leq c|x - y|^\gamma.$$

The space of Hölder continuous functions is denoted $C^{0,\gamma}(U)$.

Remark. This is "easier to show" than being Lipschitz continuous; it is a less restrictive condition than Lipschitz continuity.

Definition 3.1.2. Let $u \in C^{0,\gamma}(\overline{U})$.

- (a) The γ^{th} -Hölder seminorm is $[u]_{C^{0,\gamma}(\overline{U})} := \sup_{x,y \in U} \frac{|u(x) - u(y)|}{|x - y|^\gamma}$.
- (b) The γ^{th} -Hölder norm is $\|u\|_{C^{0,\gamma}(\overline{U})} := [u]_{C^{0,\gamma}(\overline{U})} + \|u\|_{C(\overline{U})}$.

Definition 3.1.3. The Hölder Space $C^{k,\gamma}(\overline{U})$ consists of all function $u \in C^k(\overline{U})$ for which $\|u\|_{C^{k,\gamma}(\overline{U})} < \infty$.

II. Weak Derivatives

Example 3.1.1. Suppose $u \in C^1(U)$, ∂U is C^1 , and $\varphi \in C_c^\infty(U)$. Integration by parts gives the following identity:

$$\begin{aligned} \int_U u_{x_i} \varphi \, dx &= - \int_U u \varphi_{x_i} \, dx + \int_{\partial U} u \varphi \nu^i \, dS \\ &= - \int_U u \varphi_{x_i} \, dx. \end{aligned}$$

If $u \in C^k(U)$ where $k = |\alpha|$, then by repeatedly applying integration by parts we get:

$$\int_U D^\alpha u \varphi \, dx = (-1)^{|\alpha|} \int_U u D^\alpha \varphi \, dx$$

However, if u is a function but $D^\alpha u$ does not exist in the classical sense, it is still true that the mapping:

$$\varphi \mapsto (-1)^{|\alpha|} \int_U u D^\alpha \varphi \, dx$$

from $C_c^\infty(U) \rightarrow \mathbf{R}^n$ is a well-defined linear functional. It makes sense then to define φ as the "derivative" of u in this case.

Definition 3.1.4. Let $u, v \in L_{\text{loc}}^1(U)$. We say v is the α^{th} -weak partial derivative of u , written $D^\alpha u := v$, if for all $\varphi \in C_c^\infty(U)$ we have:

$$\int_U v \varphi \, dx = (-1)^{|\alpha|} \int_U u D^\alpha \varphi \, dx.$$

Proposition 3.1.1. Let $u \in L_{\text{loc}}^1(U)$. Then $D^\alpha u$ is unique up to a set of measure zero.

Proof. Suppose $v = D^\alpha u = w$. Then for all $\varphi \in C_c^\infty(\mathbf{R})$:

$$\begin{aligned} \int_U v \varphi \, dx &= (-1)^{|\alpha|} \int_U u D^\alpha \varphi \, dx \\ &= \int_U w \varphi \, dx. \end{aligned}$$

Rearranging gives:

$$\int_U (v - w) \varphi \, dx = 0.$$

Since φ is a smooth bump function, it must be the case that $v - w = 0$ a.e., establishing the claim. ■

Example 3.1.2. weak derivative of absolute value.

Example 3.1.3. weak derivative of weak derivative of absolute value does not exist.

Definition 3.1.5. Let $k \in \mathbf{N}$ and $1 \leq p \leq \infty$. The Sobolev space $W^{k,p}(U)$ consists of all functions $u \in L_{\text{loc}}^1(U)$ whose weak derivatives $D^\alpha u$ exist and belong to $L^p(U)$. For $u \in W^{k,p}(U)$, we define its norm to be:

$$\|u\|_{W^{k,p}} := \begin{cases} \left(\sum_{|\alpha| \leq k} \int_U |D^\alpha u|^p \, dx \right)^{\frac{1}{p}}, & 1 \leq p < \infty \\ \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(U)}, & p = \infty \end{cases}.$$

For the case of $p = 2$, we redefine $H^k(U) := W^{k,2}(U)$. The closure of $C_c^\infty(U)$ with respect to the norm $\|\cdot\|_{H^k(U)}$ is denoted by $H_0^k(U)$.

Theorem 3.1.2. Let $u, v \in W^{k,p}(U)$.

- (a) For $|\alpha| \leq k$, we have $D^\alpha u \in W^{k-|\alpha|,p}(U)$. In particular, $D^\beta(D^\alpha u) = D^\alpha(D^\beta u)$ if $|\alpha + \beta| \leq k$.
- (b) $D^\alpha(u + \lambda v) = D^\alpha u + \lambda D^\alpha v$.
- (c) If $V \subset U$, then $u, v \in W^{k,p}(V)$.
- (d) If $\zeta \in C_c^\infty(U)$, then $\zeta u \in W^{k,p}(U)$ and $\frac{\partial}{\partial x_i}(\zeta u) = \zeta_{x_i} u + \zeta(D^{e_i} u)$.

Proof.



Theorem 3.1.3. $(W^{k,p}(U), \|\cdot\|_{W^{k,p}(U)})$ is a Banach space.

§ 3.2. Elliptic PDEs
